

About This Book

Conformal prediction's finite-sample coverage guarantee rests on an exchangeability assumption between calibration and test data – it is not designed, and not claimed by its own theory, to hold under distribution shift. This companion report stress-tests that assumption directly, in the same discrete-event simulation of a hospital emergency department used throughout this project, by pushing the test distribution's demand level from within the training range up to three times the training boundary.

The stress test is run twice, using two structurally different surrogate architectures: a gradient-boosting regressor, whose tree-based predictions freeze at the training boundary rather than extrapolate, and a multilayer perceptron, whose predictions keep changing as the input moves further out of range. Coverage collapses under both – but faster and more severely under the architecture capable of extrapolating, because the true relationship itself saturates under extreme demand.

An architecture's capacity to extrapolate, this report finds, is not inherently protective against distribution shift; it is protective only if the true relationship continues the trend the model learned – which it does not here.

COVERAGE AT 2X THE TRAINING BOUNDARY

0%

Multilayer perceptron – two targets, exact zero coverage

31.7%

Gradient boosting, same target, at 3x the boundary (vs. 4.7% for the other)

WHEN EXTRAPOLATION FAILS

When Extrapolation Fails

Testing Conformal Prediction's Exchangeability Assumption Across Surrogate Architectures

THE PIPELINE THIS BOOK BUILDS

